

Critical Materials: Estimating Mine-Waste Volumes

Data+

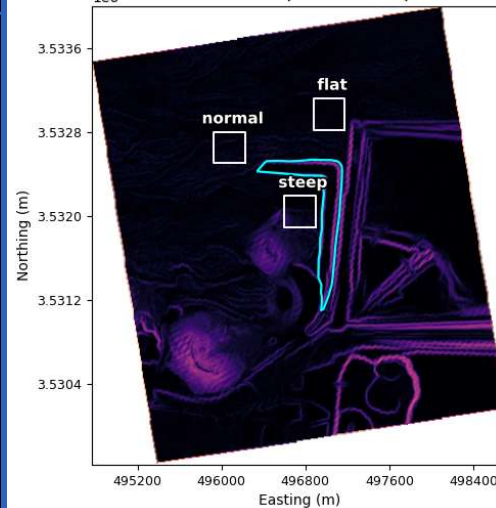
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Climate+

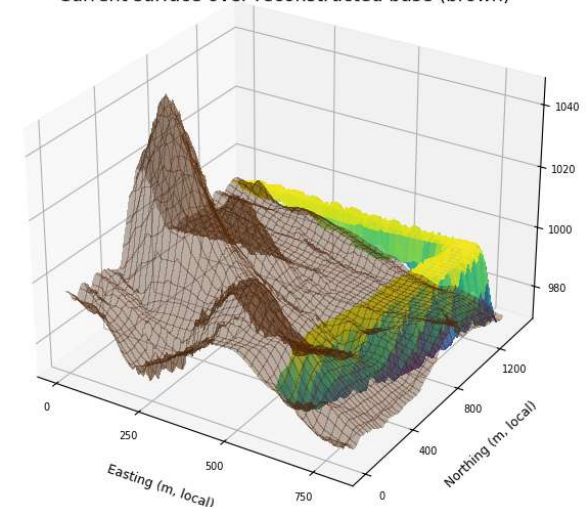
What We Need To Know And How:

- **Challenge:** Abandoned mine-waste sites may hold recoverable critical minerals, but their volumes are largely uncatalogued.
- **Benchmark:** Compare against USGS field-derived volumes for 99 sites across AZ, NC, VA, & MT.
- **Pull the dump footprint** and a surrounding buffer; download the DEM, displaying a bounding box.
- **Reconstruct the buried base surface** with a thin-plate spline anchored to the natural ground in a ring around the dump. Exclude unnaturally steep or disturbed cells.
- **To calculate volume**, subtract the floor extrapolated from the exterior of the mine feature using DEMs.
- **Validate the reconstruction** on held-out "validation squares" of known ground: steep natural, and shallow.

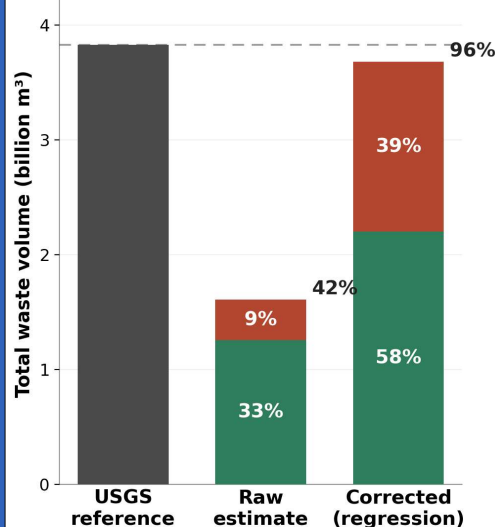
AZ000003-007 — Twin Buttes
Validation squares on slope



USGS 2,594,343 m³ • Estimate 2,619,397 m³ • +1
Current surface over reconstructed base (brown)



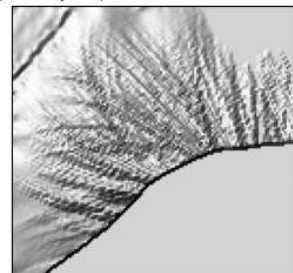
Uneven Recovery Across Different Sites



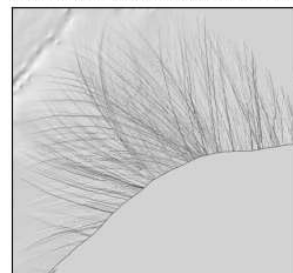
Every percentage is a share of the USGS reference total (3.83 B), so the two segments of each bar sum to that bar's recovery figure: flat sites fill 63%, mountainous fills 37%. Raw, the method recovers 42% of the reference: flat sites return 33 of their 63 points, while mountainous fills return only 9 of their 37. The regression corrects this.

AZ000015-015 — Resolution does not materially change volume estimate
USGS 16.97 M m³ • 10 m: -14.0% → 1 m: -14.8%

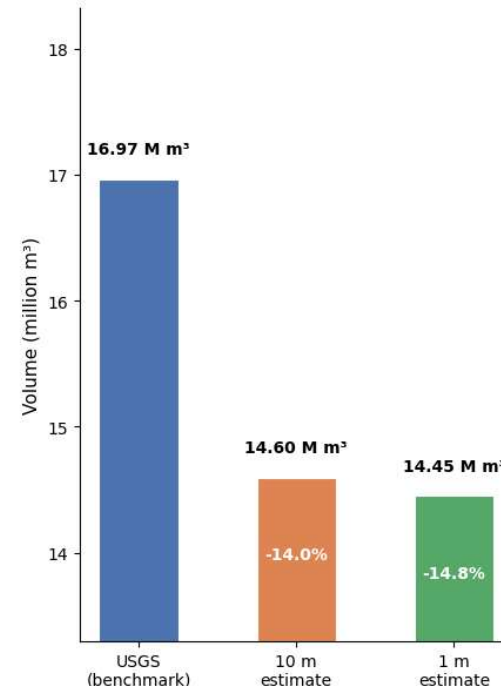
10 m (resampled): reconstructed base surface



1 m: reconstructed base surface



10 m vs 1 m resolution



What I found:

- **A DEM-only method** recovers regional waste volume to within a few percent; it *is* usable for prospecting valuation at scale.
- **Even with 5 difference spline Kernels**, the median shift to Thin Spline is 0.0%
- **The dominant error** is the buried valley floor the surface cannot reveal. Not only that, but this error is consistent across five lines of testing.
- **A control-point hook** is built and validated: the moment field exists, the method can measure those sites.
- **This project's method** is built and characterized; the control-point hook is ready to incorporate field data as it becomes available.

References

The data for this project is sourced from: USGS 3DEP, National Mine Waste Inventory. I would like to thank: Dr. Leanne Gilbertson, Clara Park, Dr. Heileen Hsu-Kim, Dr. Patrick Dunn, Dr. Anushka Mishra, and Dr. Kyle Bradbury.